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Studying social media as semiotic technology: a social semiotic multimodal framework

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ABSTRACT

How do we study social media technology? While social semiotics provides an extensive toolkit for analysing multimodal texts and semiotic practices, the study of social media as semiotic technology poses a significant challenge to existing research methodologies. In this article, we present a social semiotic framework that allows us to describe in analytical details the multimodal meaning potentials offered by digital social media technology and connect these to multimodal text-making and semiotic practices while underscoring the role of technology. Our framework is organized around seven interrelated and inherently informed dimensions: (1) multimodality, (2) practice, (3) the social, (4) medium, (5) the material, (6) the historical, and (7) the critical. This framework could pertain to most types of semiotic technologies, but will here be developed for accounting for social media technologies, and its viability will be illustrated with examples from Instagram. By developing this framework, we aim at elaborating the theoretical basis and analytical tools of social semiotics, and thereby contributing to bringing forward increased understanding of how social media technology enables making, enacting and managing meaning.

KEYWORDS

Semiotic technology; software; social practice; multimodality; social media; Instagram

1. Introduction

How can social semioticians study social media, that is, digital media that enable people to communicate and interact via virtual communities and networks? Social semioticians traditionally study semiotic resources people use for making meaning in social practices and how the use of these resources is socially regulated (van Leeuwen 2005). The central objects of study are, therefore, firstly, the *multimodal texts* of social media, such as Instagram posts, and secondly, the *social practice* that the multimodal texts are part and parcel of. However, in social media, texts are difficult to demarcate, as they appear inseparable from technological features for design, production, distribution and consumption. For instance, creating an Instagram post not only involves taking and uploading an image but also deploying resources such as image filters and hashtags. Users engage in a process in which meaning is continuously made by selecting and negotiating with a broad range of resources provided by the social medium. Therefore, a viable social

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semiotic approach to social media needs to include a third object of study: the technology with which the multimodal texts are made, and which are regulating and regulated by social practice, in other words, social media as a *semiotic technology*.

This article presents a framework for studying social media as a semiotic technology that serves to underscore our core research interest: the relationship between the multimodal meaning potentials of the semiotic surfaces of social media, the technology "below" these surfaces, and the social practices in which social media are embedded and contribute to shaping. We have found the following seven dimensions to be crucial: (1) the multimodal, (2) the practice, (3) the social, (4) the media, (5) the material, (6) the historical, and (7) the critical dimension. We will first present the research background for this framework, and then the framework itself. Finally, we will illustrate the framework with Instagram as an example. By developing this framework, we aim to elaborate the theoretical basis and analytical tools of social semiotics and thereby contributing to increase understanding of how social media technology enables the making, enacting and managing of meaning.

2. Approaching social media as semiotic technology

There is a growing number of social semiotic and multimodal discourse studies of social media texts (e.g. Zappavigna 2012; Page et al. 2014; Bouvier 2015; Adami and Jewitt 2016). Even though these studies offer valuable insights, they do not analyse the *technologies* for making and sharing social media texts. Within social semiotics the subfield of *semiotic technology* has emerged over the past few years, driven by an interest in accounting for the role of technology in meaning-making. Examples include PowerPoint (Djonov and van Leeuwen 2011, 2012, 2013; Van Leeuwen, Djonov, and O'Halloran 2013; Zhao, Djonov, and van Leeuwen 2014), Word and SmartArt (Kvåle 2016, 2018), and technologies related to specific knowledge fields, e.g. mathematics (O'Halloran 2009). So far, semiotic technology studies have included social media only to a limited degree, but important contributions include Facebook (Eisenlauer 2013), Instagram (Zappavigna 2016; Zhao and Michele 2018) and social media cross-posting practices (Adami 2014), and there are yet not any theoretical–analytical frameworks developed for approaching social media technology.

To develop such framework, this article take inspiration from the influential modelling of PowerPoint as semiotic practice and semiotic artefact, put forward by Zhao, Djonov, and van Leeuwen (2014) (see Figure 1).

The model shows that PowerPoint involves two artefacts: the PowerPoint software and the PowerPoint slideshow, connected to three semiotic practices: software designers' production of the software, users' practice of designing slideshows, and the practice of presenting with PowerPoint. The model clearly illustrates the interdependencies of technology and semiotic practice: A software designer builds the PowerPoint software, which a user/writer use to make a PowerPoint slideshow, that is then (re)used by a presenter to hold a presentation, with the semiotic artefacts of software/slides as the connecting links. The arrows and striped lines at the bottom indicate feedback loops between technologies and practices.

The PowerPoint model has prompted us to understand social media technology in a similar way in relation to social practice, to try and conceptualize this complex relationship, and provide nuanced descriptions of how social media technology becomes semiotic.

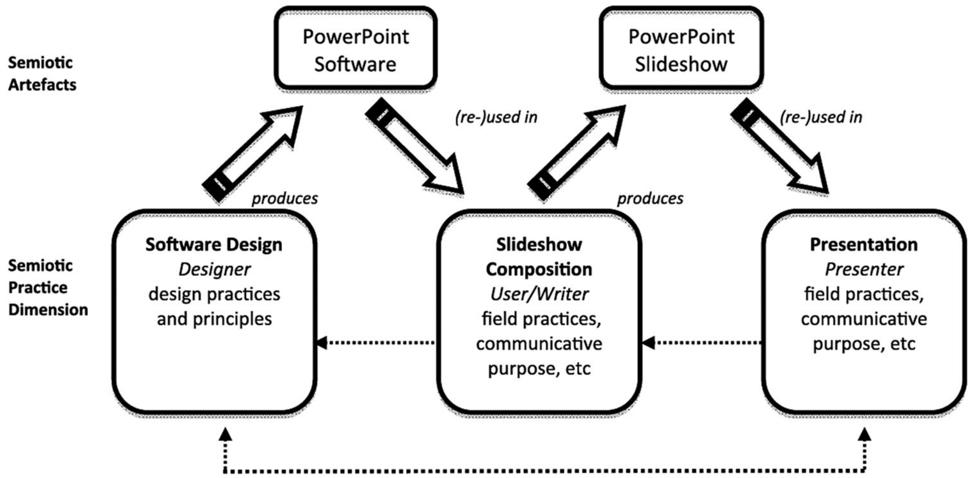

Figure 1. PowerPoint as semiotic technology and semiotic practice (Zhao, Djonov, and van Leeuwen 2014, p. 355).

Our framework is also inspired by Djonov and van Leeuwen's (2018) critical approach to semiotic software. Their approach includes three dimensions: the design of software products, their use in social settings, and the relationship of software design and use to broader semiotic and cultural practices. Our article also takes a holistic approach to social media technology, but our proposed framework modifies their three dimensions into seven to make them analytically usable for approaching social media as semiotic technology.

3. A social semiotic framework for studying social media technology

Our framework's proposed object of analysis is the design and use of social media technology for making multimodal texts in social practices. Figure 2 shows that it takes departure in the triad of multimodal texts, social practice, and semiotic technology, as mentioned in the introduction, and proposes that this triad is to be understood from seven interrelated and mutually informing dimensions: (1) the multimodal, (2) the practice, (3) the social, (4) the media, (5) the material, (6) the historical, and (7) the critical dimension.

The framework treats "semiotic technology" not merely as a "tool" that serve some pragmatic purpose but as an artefact that structures how people communicate and interact. To elaborate on the function of this artefact, we adopt the Italian thinker Agamben's (2009) conceptualization of technology as "apparatus," that is, a material-ideological structuring of knowledge and action which "has in some way the capacity to capture, orient, determine, intercept, model, control, or secure the gestures, behaviors, opinions, or discourses of living beings" (Agamben 2009, 14). The proposed framework thereby seeks to bring forward the analytical and theoretical approach to social media technologies.

3.1. The multimodal dimension

Multimodal analysis is the hallmark of social semiotics. The multimodal dimension concerns systematic and detailed analysis of the semiotic *resources* and semiotic

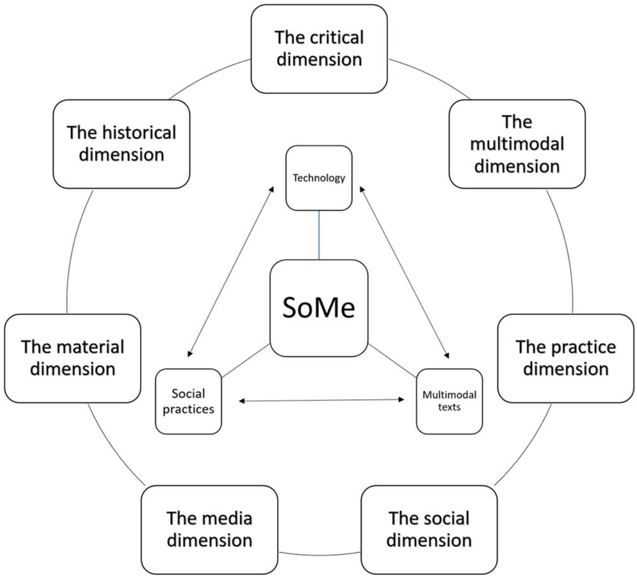

Figure 2. Framework for the study of social media as semiotic technology.

regimes that people use and attend to in social practices. Van Leeuwen (2005, 3) defines semiotic resources as "the actions and artifacts we use to communicate, whether they are produced physiologically – (…) – or by means of technologies – with pen, ink and paper; with computer hardware and software (…)." Semiotic regime refers to the social mechanisms that regulate the use of semiotic resources. Van Leeuwen (2005, 2008a) addresses several kinds, including codification (explicit rules), tradition (implicit rules), expertise, role models, and technology, that all structure production and interpretation of meaning.

Extending the study of multimodality to social media technology means describing digitally mediated resources in social media, such as writing, emojis, and gifs, and their meaning potentials. Thus, the technical features for making and sharing content are considered semiotic resources, in terms of how they enable people to communicate. Social media is, like any software, similar to, yet different from other semiotic systems. Importantly, all resources in digital technologies have been deliberately designed (Djonov and van Leeuwen 2012). The design and organization of resources in the layout of a social medium are often impossible to modify for the individual user, as software designers pre-determine its composition. Thus, the display of selected resources (as well as other possible resources not selected) is a vital focus of the multi-modal analysis.

Semiotic regimes are rules for semiotic practice and can be built into a social media technology. This echoes Agamben's description of the structuring function of an apparatus. Drawing on Van Leeuwen's (2005, 2008) account of regimes, it follows that a given social media technology can function as a regime in itself, as it structures a semiotic practice. The other regimes that van Leeuwen describes may also be built into a specific social media technology and regulate the design, use, and discourses of that technology (Roderick 2016).

3.1. The practice dimension

Multimodality studies investigate semiotic resources and regimes in various social *practices*, i.e. "socially regulated ways of doing things" (Van Leeuwen 2008b, 6). Social media are developed and produced in specific practices, and people use them for carrying out various social practices in their everyday and professional lives.

The connections between a social media technology and a social practice are in some cases quite specific, for instance between dating and Tinder, but are in other cases, for instance, Facebook, more complex (Brügger 2015). Facebook has exceeded far beyond the practice of student life in the dorm rooms of Harvard and is today used in a broad variety of social practices, e.g. advertisements, family contact, birthday greetings and political debates. Thus, entering a social medium environment like Facebook involves entering a network of social practices that are dependent on the medium in various ways and to various degrees.

In some cases, social media technology is a necessary means for carrying out all stages of a social practice, while other social practices may involve social media only in some stages. Social media technology can also contribute to reconfiguring the actions of a practice and their order. For instance, before YouTube, one would read books and perhaps attend a class to learn advanced cooking techniques. Today, people might search for and watch a YouTube video on specific techniques, and later post photos of their food experience on social media. Thus, relating social media to practice is not just a matter of connecting a specific technology to a specific practice and analysing to what extent that technology is part of the enactment of the practice, but also of how the technology resemiotizes the practice and contributes to changing the practice itself (Iedema 2003). Thus, this dimension of the framework seeks to uncover how social media technology brings about transformations of social practices.

3.3. The social dimension

Amongst the fundamental assumptions in social semiotics is the idea that language and other semiotic systems are intrinsically *social* (Halliday 1978). In a traditional social semiotic approach, one can distinguish between three levels of the social:

1. The social as the basis for all meaning-making: All semiotic work and processes are fundamentally social, as the social is generative for meaning-making; "the source, the origin and the generator of meaning" (Kress 2010, 54).
2. The social as the social context: The social environment, i.e. the situational context where social actors communicate, which is part and parcel of a broader socio-cultural historical context.
3. The social as a strand of meaning: Meaning is inherently social and analysed as three strands of meaning that operate together to serve different functions in a social context, in other words, in terms of ideational, interpersonal, and textual metafunctions.

These three intrinsically related levels can be analytically operationalized for approaching social media as semiotic technology. Firstly, by *studying social media technology from a social perspective* (cf. point 1), i.e. connecting semiotic activities performed in social media to social, cultural, economic and political factors that influence practices, resources and communication technologies. It includes analyses of characteristics of social media in relation to, for instance, the global reach of social media, user-generated content, many-to-many participation, accessibility, connectivity, mobility, convergence, or in other words; to view social media as semiotic responses to social conditions and factors in late-modern societies (Kress 2010, 18–31). Other important dimensions are how social and economic factors shape identity construction, social organization, and power struggles within the techno-cultural infrastructure of social media.

Secondly, the social dimension involves *studying social media technology as part of social contexts*, i.e. investigating how people use social media in specific social contexts (cf. point 2). Technology is part of this dimension in terms of how it contributes to structuring, facilitating and adjusting social environments and their elements in relation to a practice (cf. Agamben's concept of technology). One example is how the move from face-to-face interactions in university classrooms to teacher-monitored Facebook groups, frames the social context of discussion, feedback and assessment differently.

Thirdly, the social dimension of the social media technology be *analysed in terms of the metafunctional meaning potential* that a social media technology is designed to create (cf. point 3). While social media technologies afford resources for making ideational, interpersonal and textual meaning, we propose that the "social" of social media can be specifically related to the interpersonal metafunction and that social media is distinctive by providing an extended palette of resources for interpersonal meaning-making. Compared to so-called old media, social media are often regarded as offering new and/or broader sets of resources and principles for social interaction (Klastrup 2016). This dimension implies that a possible answer to the question of what the *social* in social media involves is that social media facilitate a broader range of possibilities for interpersonal meaning-making than traditional media. However, different social media may facilitate different degrees of sociability qua the interpersonal semiotic resources they provide.

In addition to the three traditional levels addressed above, we will add a fourth level: *the technical aspects of social media as social expressions.* This dimension involves examining how social media as technical constructs incorporate social ideas and norms about information, communication, social interaction and organization (Langlois 2014; Helmond 2015). One (in)famous example is the change in the sociability of Instagram's news feed algorithm in March 2016. The new algorithm no longer organized posts in reversed chronological order but based on the social interaction around posts. The algorithm that technically supports this feature facilitates a view on being social that is modelled on *popularity*. Drawing on Agamben's notion of apparatus, the technology that highlights the popular posts to a large extent also structures the social interactions on the social medium. The social media system awards this particular kind of interaction and prompts users to take part in this social activity. The new algorithm also supports social mechanisms that strengthen the social standing of already popular individuals (e.g. celebrities and politicians) with cultural and social capital (Bourdieu 1979).

3.4. The media dimension

In social semiotics, "media" refers to the material resources used to realize meaning in social practice (Kress 2010), and is seen as the material substance of semiotic resources. One can distinguish between *production media* and *distribution media*. The former is seen as "resources in the production of messages" (Kress and van Leeuwen 2001, 22) and includes "both the tools and the materials used" (e.g. a chisel and a block of wood), while distribution media refers to "resources for the dissemination of meanings as message" (Kress 2003, 23). Production media may further be divided into three kinds of technologies: (i) "technologies of the hand," i.e. representations made by the human hand and/or hand-held tools; (ii) "recording technologies," i.e. tools that make analogue representations of material articulation of a semiotic events such analogue photography and film; and (iii) "synthesizing technologies," i.e. tools that make digitalized representations (of handmade or analogue representations) via an interface (Kress and van Leeuwen 2006, 217). Social media may include both production and distribution media. A description of semiotic resources (cf. section 3.1) may therefore also include a description of the resources for the material production and distribution of meaning that particular social media afford. The study of digital distribution resources is particularly interesting, since a distinctive feature of social media is the distribution of meaning in networked organizations of social media participants. Jenkins, Ford, and Green's (2013) study of "spreadable media" has shown how content spreads in new digital media, while Walton et al. (2012) has shown different kinds and degrees of phone and mobile media sharing, but more work needs to be done in this field.

Social media are digital media, thus to be categorized as synthesizing technologies, but it is still relevant to acknowledge their relations to "technologies of the hand" and "recording technologies." Studies of how social media remediate previous media technologies can, for instance, help us understand the formation of contemporary mediascapes (Appadurai 1996) and how social media function as production media for semiotic work in various social settings (cf. section 3.2). When social media are seen as synthesizing technologies, software becomes important (Djonov and van Leeuwen 2018). Even though Kress and van Leeuwen's (2001) description of production media includes both tools (e.g. a software program) and materials (e.g. photos), which is still a useful distinction, it is vital to acknowledge that the tools and materials of social media consist of the same digital "substance," namely software. A leading scholar of Software Studies Lev Manovich (2013) describes software as a medium that "simulates/virtualizes previously existing physical or electronic media and their techniques for generating, editing and accessing content – plus adds new properties" (337). The "adding of new properties" is of profound importance, as moving old media into new software media by way of digitalization not only means that meaning is remade when moved across media and modes (glossed "transduction" by Kress (2010)) but also that the resources of modes themselves can be transformed in terms of the meaning potentials they afford when represented digitally in a software medium. To borrow Manovich's example, an image has different affordances (e.g. dynamic and interactional features) depending on the kinds of software (and hardware) used to access and display the image, which are different from the meaning potential of that same image in an analogue medium. Social media software analysis with special emphasis on how software reconfigures resources is, therefore, an important part of the media dimension in the proposed framework.

3.5. The material dimension

According to Kress (2010, 105), a social semiotic approach emphasizes "the *material*, the *physical*, the *sensory*, the *bodily*, the 'stuffness of stuff', away from abstractions, towards the specific, the variable." Materiality matters – it matters to how and what kind of meanings are made, also in social media settings.

One aspect of the materiality of social media concerns the physical materiality of the device. The primary devices for social media are desktop computers, tablets, and importantly, mobile phones. The sheer size of the device is often connected to how social media are used, for instance, the small smartphone screen renders it a *private* medium, while a 25-inch computer screen allows other to watch or take part. The mobility and quite private screen of mobile phones also enable the extension of social media activities into simultaneous offline social practices, and vice versa, as well as the simultaneous participation in *other* social practices. The small size of smartphones thus allows for a broad spectrum of social interconnections as well as shifting attention between the various online and offline interactions.

Also, devices have a physical impact on society and the world, which ties into the broader discussion of the politics of digital materialism (Leonardi 2010; Casemajor 2015) and the real-life consequences and concerns that digitalization brings along, including e.g. ownership, laws and regulations, surveillance, localization of data storage, digital pollution and environmental issues of data waste (Bennett 2009; Parikka 2012; Fuchs 2014).

Furthermore, codes and algorithms are material manifestations of software. Codes are here used as "a general term for a wide variety of different concrete programming languages and associated practices" (Berry 2011, 33) that make up the digital substance of software, including social media. The semiotic work of software production is informed by and inscribes social norms into the codes and creates links between physical and digitally represented materialities. Codes make up the digital "stuff" of social media; they afford the properties, parameters and rules that social media technologies provide their users with. While end-users do not directly see the codes, they do nevertheless interact with and experience them as coded digital materialities whenever they create and share content.

It is uncontroversial to claim that codes and algorithms matter. However, most researchers in the humanities and social sciences are not trained in coding and computation and cannot access and make sense of the technical infrastructure – and in most social media, codes are not available to users. One possibility is, therefore, as Djonov and van Leeuwen (2018) do, to exclude algorithms and codes from analyses of "software for making meaning" (567). This is a pertinent methodological choice, but social media technology studies might also address codes more directly; by interrogating how, to whom and at what point in the meaning-making practice a technology becomes semiotic. For instance, the same digital semiotic technology can provide different affordances to different sign-makers, i.e. software designer and end-user. Such interrogations may also provide a response to potential criticism of "screen essentialism" (Waldrip-Fruin 2009, 3; quoted in Berry 2011, 36), i.e. studies of software that stays at the level of the user interface without accounting for the computational levels of information processing "below" the screen surface which make the digital mediation of semiotic resources such as writing, photos and music available.

Since codes and algorithms are the material used to produce social media software, they can in themselves be considered semiotic resources. The semiotic work performed by codes and algorithms involves a recontextualization of staged-goal-oriented actions, and algorithms are thus essentially *generic* structures, and the rich tradition of genre studies in social semiotics (Martin 1992; Van Leeuwen 2005) can, therefore, be drawn on for addressing how codes realize functions that serve a purpose in relation to a social practice that the computational processes are part off. Even though codes and computations are, or appear to be, incomprehensible, they are ultimately modelled on social practices. A similar line of approach can be found in digital humanities (e.g. Berry 2016; Bucher 2016) and in critical code studies (e.g. Marino 2006).

Finally, the material dimension concerns how materialities are *represented* in various social media. Like the other dimensions, this is not necessarily specific to social media studies. Previous semiotic technology studies have for instance shown how analogue materials like fabrics are represented digitally in PowerPoint slide templates (Djonov and van Leeuwen 2011), and how tactile qualities of clothes are resemiotized when shopping moves from the physical store to online shops (Andersen and van Leeuwen 2017). On the digital surface of a social medium, e.g. a photo app, represented materialities may also inscribed into the predesigned templates and tools of the app, for instance resources for representing "fainted" photos, scratched surfaces and other material properties from different camera and film types (Poulsen forthcoming).

3.6. The historical dimension

A historical perspective of semiotic resources is integral to social semiotics. To Van Leeuwen (2005), a diachronic description means examining how resources come into being and undergo changes when people need or want new ways of using existing resources and/or invent new resources. Such semiotic changes reflect and shape changes in society (Kress 2010).

Studies of semiotic technologies other than social media have demonstrated the value of a historical perspective. Examples include changes in mathematical notations throughout history (O'Halloran 2009), texture and layout in various versions of PowerPoint (Djonov and van Leeuwen 2011), and visual hierarchies in different versions of Word (Kvåle 2016). Despite the recent emergence of social media technologies, important changes in their design and use can already be observed.

One aspect of this dimension concerns the *historical development of a given social medium* – previous versions, update history, added features, withdrawal of features, and so on. Inspired by media archaeology (e.g. Huhtamo and Parikka 2011), investigations of such developments can reveal the constructions of various, potentially conflicting, histories of a medium, thereby underscoring the complexity of a social medium environment's dynamics and its connections to different interests of social actors and power struggles among them (Kress 2010). Such investigations could also offer valuable insights into how a social media technology enters into a long tradition of *previous media*. This aspect concerns how social media takes inspiration from and remediates and alternates the properties and characteristics of "old" media (Bolter and Grusin 1999). The historical interest also concerns the role of social media in transformations of existing social practices and the emergence of new practices. The historical perspective enables linking technology to practices and showing how a technology grows out of particular practices (cf. section 3.2).

Finally, the historical dimension includes the *discourses that surround social media* – the ways a social medium is introduced, marketed, discussed and criticized, as well as the ways users adopt, adapt or reject them. Many social practices that include social media are deeply embedded in normative discourses on how actions should be performed, how actors should style and present themselves, and on how the setting should be arranged (Djonov and van Leeuwen 2012; Roderick 2016).

3.7. The critical dimension

Social semiotics and multimodality studies are closely related to critical discourse analysis (CDA) and critical theory in general (Fairclough 2003). CDA studies how discursive practices are shaping social practices and vice versa, with an attention to social wrongs: "the way social power abuse, dominance and inequality are enacted, reproduced, and resisted by text and talk in the social and political context" (Van Dijk 2001, 352).The are many versions of CDA, and we are in particular inspired by the Discourse-Historical Approach (DHA) which concerns three interrelated aspects: (i) Text or discourse immanent critique that seeks to map inconsistencies, (self)-contradictions, paradoxes and dilemmas; (ii) socio-diagnostic critique that aims to uncover the persuasive or "manipulative" strategies in discursive practices, and iii) prospective critique that seeks to improve people's communication skills and empower social actors with better resources for participation (Wodak 2015, 24–25). We also seek to broaden the critical scope by also including design and usages of social media technologies. Such studies already exist in the field of semiotic technology, like the critique of marketization in PowerPoint software (Djonov and van Leeuwen 2014), neoliberal management discourses in Word and SmartArt (Kvåle 2016), the promotional aesthetics in website design (Kvåle and Poulsen forthcoming), and visual discourses on Instagram's corporate blog (Poulsen 2018a). The role of social media technologies in contemporary society calls for continued research and critique.

The critical perspective on software is a standalone dimension of the framework, but it could also be included in the other dimensions. For instance, the multimodal dimension could include a critique of the preselected resources and built-in constraints chosen and designed by social media developers, and the practice dimension could involve critical discussion of how social media platforms let companies assemble users' personal data in order to elaborate their marketing techniques (Serazio and Duffy 2018). The changing practices of digital marketing and the distribution of advertisements in software and non-human actors are central fields for critical semiotic technology studies, whether negatively by pointing out "how exploitation, domination, commodification and ideology interact in shaping media communication in society" (Fuchs 2014, 255), or positively by investigating "what potentials for alternatives there are, and how struggles can use and advance these potentials" (255).

4. Example: Instagram as a semiotic technology

We will now apply the framework to Instagram, with a focus on how the design of the mobile app user interface provides users with a technology for making, editing and sharing photos. The example is not intended as an exhaustive analysis, but to illustrate the framework (Table 1).

The multimodal dimension: In the mobile user interface, the main sites of display are the personal profile site, the feed, and the various editing displays. Table 1 presents a

Table 1. Semiotic analysis of selected tool in the Instagram image editing interface.

Technical function		Semiotic function		
Name of tool	What the tool does	Semiotic representation	Tool in semiotic terms	Meaning potential
Selected tools for picture making/recording in Instagram, version 2.0				
Take photo	Records photo with mobile camera	Camera icon Blue circle w. dotted line	Existential process	Ideational: the act of representing something as "being there" (Boeriis 2009: 180)
Cut and Scale	Crops and resizes photo, following Rule of thirds, into square format	Grid / two horizontal lines and two vertical (9 squares)	Centre-margin, Left-Right Top-Down + (Existential process)	Textual: Information value
Tools for picture editing				
Rotation & zoom	Rotates the photo 360 degrees, in increments of 90 degrees. In addition, lets one tilt the photo 25 degrees to the right or left while zooming in	Square with dotted line	Lateral point of view Distance + (existential process)	Interpersonal: Position of the viewer
Frame	Puts a frame around the photo	Square embedded in another Square	Segregation (Van Leeuwen 2005)	Textual: Framing
Tilt shift	Changes focus i photo. Blurs out Two ways: linear or circular focus	White drop	Focus contrast	Textual: Salience
Lux effect	Makes the photo lighter or darker	Black/white circle with array	Light intensity and contrast (modality marker) Tonal contrast	Interpersonal: Modality Textual: Salience
Filters	Adds light and colour effects	Picture of balloon and name of filter	Depends on the filter, e.g. Lo-fi: Colour saturation/ light intensity + Colour contrast+ Frame	Interpersonal: Modality Textual: Salience and Framing
Selected tools for picture sharing				
Photo map	Adds data about where the photo was taken	Text + slider Diagrammatical map	Locality	Ideational: Circumstance
Image text/Hashtag	Describes photo and makes it searchable	White text field	Elaboration/Extension	Ideational: Image–text relations
Sharing on social media	Distributes photo on social media networks	Logo & name of social media + slider	Speech function: Offer (to share photo in different ways)	Interpersonal: Information exchange

description of technical and semiotic functions of some of the tools in the editing display in version 2.0. The technical function describes the name of individual tools and what they do, e.g. rotate the photo or shape as a square. Under the semiotic function, each tool is seen as affording a set of semiotic resources that users can apply. The description includes how the tool is digitally represented in the user interface, such as a water drop or a slider, followed by a semiotic term for what the tool does, for example, increase colour brightness, and finally, its dominant metafunctional meaning potential. Some tools work as simple resources, e.g. for changing colour saturation, while other tools incorporate a bundle of resources, such as the filters' resources for modifying colours, frames, and focus (Poulsen forthcoming). Thus, the multimodal dimension demonstrates which resources for meaning-making are built into the Instagram technology, and how.

Furthermore, the Instagram user interface design comes with certain semiotic regimes. Inspired by Djonov and van Leeuwen's (2012) approach to the normativity of semiotic software, we see the editing display of Instagram as containing both paradigmatic and syntagmatic regulating mechanisms. The paradigmatic mechanisms in the mobile app allow scalable uses of specific resources, e.g. degrees of colour saturation, as well as binary uses, e.g. application of framing devices. Also, some photo filters are foregrounded and thereby prompt users to select them, while other (older) filters first must be located and activated to be available. Likewise, certain visual features are preselected when a photo is captured, for example the "Add Location" and the square photo format, thereby forcing the user to make an active choice if s/he does not want these features included in a post.

The syntagmatic semiotic regimes of Instagram structure the workflow, i.e. the order of photo editing processes. For instance, Instagram only lets users apply filters after, not before a photo is taken, in contrast to the Hipstamatic app (Halpern and Humphreys 2016). Additionally, in the first versions of the Instagram app, the number of resources available was more limited, and the order in which visual semiotic resources could be combined was more restricted than in newer versions. Filters were previously "locked" in the design, giving users no options to decide on the filters' modifications. After version 6.0, it became possible for the user to apply a filter to various degrees. However, the possible modifications of an Instagram photo are nevertheless restricted. The technology preserves basic photographic features and ensures that photos end up looking "good," so resources are still structured in a way that "secures" the aesthetic qualities of an edited photo (for further description, see Poulsen forthcoming).

The practice dimension: The practice of social photography depends on technologies such as Instagram. Instagram merges at least three stages of photographic practices: photo making, photo processing, and photo showcasing. Historically, these practices were kept apart and required specific knowledge and skills, but with Instagram and similar technologies they are resemiotized and merged into one and become accessible to a broad community of amateur photographers. For this process, Instagram's software designers have selected some actions from each practice and built some photo processing techniques into the editing toolbar.

Instagram has also given rise to new social photography practices. One local example is the "Instameet" at Copenhagen Airport in 2014, where a group of Instagram users were invited on a photo expedition to restricted areas of the airport, and afterwards hashtagged (#copenhagenairport_instameet) and shared their photos on Instagram. The airport used this event in their promotion strategy, and thereby included the Instagram technology in the practice of corporate communication.

The social dimension includes several aspects, and we will here, in continuation of the multimodal dimension, focus on how Instagram provides an extended palette of interpersonal resources. Instagram's palette includes for instance resources for visual modality, e.g. colour saturation and focus, which can function as markers of kinds and degrees of "truth" and "reality" (Kress and van Leeuwen 2006). Another important visual interpersonal resource is the filters. Many filters appear to intend to evoke feelings of nostalgia, and the use of a filter thereby becomes a way to orchestrate certain social encounters between an image and the viewer.

Instagram's palette of interpersonal resources also includes resources for showcasing, which can establish various kinds of photo sharing encounters. The Instagram technology allows users to share their photos either on Instagram's own social network platform or on certain other social media platforms such as Facebook or Twitter. Different platforms permit different kinds of interactions and thereby provide different interpersonal meaning potentials. Instagram prompts a rather emotional response through "hearts," emojis and a visually based dialogue. In this way, the technology explicates and amplifies interpersonal aspects of communication, and frames social media interaction as a kind of a management whereby the user as an administrator selects and decides what kind of users and feedback possibilities she want to engage with (albeit the sharing of photos is beyond a single user's control). Thus, Instagram becomes a semiotic technology not only for making meaning but also for managing the exchange of meaning.

In *the media dimension*, we concentrate on two aspects: the media tradition and Instagram as a software medium. Instagram is known for taking inspiration from older media technologies such as analogue, Instamatic cameras, which are remediated primarily through filters, e.g. *1977* or *Kelvin*, but also other media and their features are digitally remade in the technological design. Compared to a media studies approach to remediation of older, especially analogue, photo media and materials (e.g. Caoduro 2014), a semiotic technology analysis focuses on the specific resources for representing content, styles and aesthetics of other media (see Poulsen forthcoming).

Furthermore, following Manovich's claims about software as a medium, Instagram can be seen as simulating other (older) media and adding new features. To give an example, in May 26, 2011, Instagram introduced the possibility of "liking" a photo by double-tapping it, thereby adding an interactive resource that had not yet been part of photos in analogue media. This digital mediation enabled a meaning potential that invites a more direct physical interaction with photos that affords a positive appraisal. In this way, the palette of interpersonal resources in the user interface is reconfigured by Instagram as a software medium.

For *the material dimension*, we will elaborate on the *represented materialities* in the Instagram app. The design of photo filters and other resources in the user interface often take inspiration from the materiality of analogue, Instamatic photography, as already mentioned. The design of photo filters also draws inspiration from photographic materials, (chemical) processes, such as cross processing, and techniques, for instance, tintypes and vignettes. On this basis, Poulsen (forthcoming) argues that photo filters, on the one hand, get their meaning potentials from a history of picture making with diverse photographic tools and materials, and on the other hand, erase the very same distinct cultural references when these materials are digitally represented in the software medium of Instagram. The filters thereby depart yet are deprived of their cultural heritage and meaning potentials, and the use of represented materialities must adapt to an "Instagram aesthetics" that serves first and foremost to make photos "beautiful."

The *historical dimension of Instagram* includes the design history of photo editing and sharing tools in Instagram's graphical user interface. Three changes can be singled out: the number of tools, the representations of tools, and the semiotic function of tools (Poulsen 2018b). The number of tools changed significantly especially in the first two years of Instagram's history (2010–2012), which presumably reflects a period of adjustment and optimizing of the user interface design based on end-users' use and feedback. In the first years, the app made available a limited number of filters that primarily adjusted and compensated for poorly shot photos; later, the number increased, reportedly to meet a demand by users for more tools for visual experiments and creative expression.

The representation of tools in the interface has also changed from strongly inspired by analogue and Instamatic camera functions to a more generic look. The changes in tool and social button design also reflect trends in web design; from skeuomorphic design (physical object resemblance) to minimalistic flat design. Changes in the semiotic function of these tools also reflect social changes and needs for new (visual) resources (Kress 2010). We have already mentioned the function of double-tapping, and another example is the move from binary application (on/off) of filters to gradual filters, which enable users to make subtler visual meanings.

The *critical dimension* concerns social values, politics and ideology of Instagram in terms of its design, use and discourses, and here we will focus on the normative discourses on the meaning potentials of the tools. A critical analysis of Instagram's corporate blog posts about tools showed that the corporate blog used user-generated images to represent a (super-)naturalistic, yet sensual outside world. In doing so, "the blog facilitates an aesthetics of 'things' that are singled out and presented as beautiful, stunning and amazing in their mere appearance" (Poulsen 2018a). In these images, people are almost entirely absent. They show no social activities and depict no social engagement. Thereby, the images enact a neutral point of view that positions the blog viewer as a passive observer who is invited to simply gaze at the presented outside world. A text immanent critique (see section 3.7) of this discourse is that it departs from the acclaimed company mission of Instagram to enable people to be social and share their photos. This also implies a socio-diagnostic critique of the way that the corporation includes the images made by Instagram users to justify and promote changes in the app design within the practice of corporate blogging. The rhetorical strategy seems to be that because the Instagram community have started using newly launched tools in their photographic practices, these tools are approved by users, and the Instagram Corporation then simply puts users' practices forward as factual reports about how the tools enable users to make better images.

5. Conclusion

In this article, we have presented a social semiotic framework for studying social media as semiotic technology. This framework allows for analytical descriptions of the multimodal meaning potentials of social media connected to social practice, while critically acknowledging social media technologies as material, historical, and social constructs. The breadth of this framework means that several dimensions still needs more work. We will in future research projects try and address these, and it is our hope and aim that other scholars will contribute by elaborating, critiquing, and further developing the approach to social media as semiotic technology.

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No potential conflict of interest was reported by the authors.

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References

Adami, Elisabetta. 2014. "Retwitting, Reposting, Repinning; Reshaping Identities Online: Towards a Social Semiotic Multimodal Analysis of Digital Remediation." *LEA: Lingue e Letterature d'Oriente e d'Occidente* 3 (3): 223–243.

Adami, Elisabetta, and Carey Jewitt. 2016. "Special Issue: Social Media and the Visual." *Visual Communication* 15 (3): 263–270.

Agamben, Giorgio. 2009. *What is an Apparatus? And Other Essays*. Stanford, CA: Stanford University Press.

Andersen, Thomas Hestbæk, and Theo van Leeuwen. 2017. "Genre Crash: The Case of Online Shopping." *Discourse, Context & Media* 20: nov, 191–203.

Appadurai, Arjun. 1996. *Modernity At Large: Cultural Dimensions of Globalization*. Minneapolis: University of Minnesota Press.

Bennett, Jane. 2009. *Vibrant Matter: A Political Ecology of Things*. Durham, NC: Duke University Press.

Berry, David M. 2011. *The Philosophy of Software: Code and Mediation in the Digital Age*. Basingstoke: Palgrave Macmillan.

Berry, David M. 2016. "The Post-Archival Constellation: The Archive Under the Technical Conditions of Computational Media." In *Memory in Motion. Archives, Technology, and the Social. Recursions: Theories of Media, Materiality, and Cultural Techniques*, edited by Ina Blom, Eivind Røssaak, and Trond Lundemo, 103–125. Amsterdam: Amsterdam University Press.

Boeriis, Morten 2009. Multimodal Socialsemiotik & Levende Billeder [Multimodal social semiotics and moving images]. Unpublished PhD thesis. Department of Language and Communication. The Faculty of Humanities. University of Southern Denmark.

Bourdieu, Pierre. 1979. *Distinction: A Social Critique of the Judgment of Taste*. London: Routledge.
Bouvier, Gwen. 2015. "What Is a Discourse Approach to Twitter, Facebook, YouTube and Other Social Media: Connecting with Other Academic Fields?" *Journal of Multicultural Discourses* 10 (2): 149–162.
Bolter, Jay David, and Richard Grusin. 1999. *Remediation: Understanding New Media*. Cambridge, MA: MIT Press.
Brügger, Niels. 2015. "A Brief History of Facebook as a Media Text: The Development of an Empty Structure." *First Monday* 20 (5), Accessed: 20 January 2018. Available at: http://firstmonday.org/ojs/index.php/fm/article/view/5423/4466.
Bucher, Tania. 2016. "Neither Black Nor Box: Ways of Knowing Algorithms." In *Innovative Methods in Media and Communication Research*, edited by Kubitschko Sebastian and Kaun Anne, 81–98. Cham: Palgrave Macmillan.
Casemajor, Nathalie, 2015. "Digital materialisms: Frameworks For Digital Media Studies." *Westminster Papers in Culture and Communication* 10 (1): 4–17.
Caoduro, Elena. 2014. "Photo Filter Apps: Understanding Analogue Nostalgia in the New Media Ecology." *Networking Knowledge* 7 (2): 67–82.
Djonov, Emilia, and Theo van Leeuwen. 2011. "The Semiotics of Texture: From Tactile to Visual." *Visual Communication* 10 (4): 541–564.
Djonov, Emilia, and Theo van Leeuwen. 2012. "Normativity and Software: A Multimodal Social Semiotic Approach." In *Multimodality and Practice: Investigating Theory-in-Practice-Through-Method*, edited by Sigrid Norris, 119–137. New York: Routledge.
Djonov, Emilia, and Theo van Leeuwen. 2013. "Between the Grid and Composition: Layout in PowerPoint's Design and Use." *Semiotica* 2013 (197): 1–34.
Djonov, Emilia, and Theo van Leeuwen. 2014. "Bullet Points, New Writing and the Marketization of Public Discourse. A Critical Multimodal Perspective." In *Critical Multimodal Studies of Popular Discourse*, edited by Emilia Djonov and Sumin Zhao, 232–250. London: Routledge.
Djonov, Emilia, and Theo van Leeuwen. 2018. "The Power of Semiotic Software: A Critical Multimodal Perspective." In *The Routledge Handbook of Critical Discourse Analysis*, edited by John Flowerdew and John E. Richardson, 566–581. London: Routledge.
Eisenlauer, Volker. 2013. *A Critical Hypertext Analysis of Social Media: The True Colours of Facebook*. London: Bloomsbury Academic.
Fairclough, Norman. 2003. *Analysing Discourse. Textual Analysis for Social Research*. New York: Routledge.
Fuchs, Christian. 2014. *Social Media – A Critical Introduction*. Los Angeles, CA: Sage.
Halliday, M. A. K. 1978. *Language as Social Semiotic: The Social Interpretation of Language and Meaning*. London: Edward Arnold.
Halpern, Megan, and Lee Humphreys. 2016. "Iphoneography as an Emergent art World." *New Media & Society* 18 (1): 62–81.
Helmond, Anne. 2015. "The Platformization of the Web: Making Web Data Platform Ready." *Social Media+Society*, July–December, 1–11.
Huhtamo, Erkki, and Jussi Parikka, eds. 2011. *Media Archaeology: Approaches, Applications, and Implications*. Berkeley, CA: University of California Press.
Iedema, Rick. 2003. "Multimodality, Resemiotization: Extending the Analysis of Discourse as Multi-Semiotic Practice." *Visual Communication* 2: 29–57.
Jenkins, Henry, Sam Ford, and Joshua Green. 2013. *Spreadable Media: Creating Value and Meaning in a Networked Culture*. New York: New York University Press.
Klastrup, Lisbeth. 2016. *Sociale netværksmedier* [Social Network Media]. Frederiksberg: Samfundslitteratur.
Kress, G. 2003. *Literacy in the New Media Age*. London: Routledge.
Kress, Gunther. 2010. *Multimodality. A Social Semiotic Approach to Contemporary Communication*. London: Routledge.
Kress, Gunther, and Theo van Leeuwen. 2001. *Multimodal Discourse. The Modes and Media of Contemporary Communication*. London: Arnold.

Kress, Gunther, and Theo van Leeuwen. 2006. *Reading Images – The Grammar of Visual Design*. 2nd ed. London: Routledge.
Kvåle, Gunhild. 2016. "Software as Ideology. A Multimodal Critical Discourse Analysis of Microsoft Word and SmartArt." *Journal of Language and Politics* 15 (3): 259–273.
Kvåle, Gunhild. 2018. "Expressing Identity in Microsoft Word. A Critical Discussion of the Stylistic Normativity of Templates and Software." In *The Materiality of Writing. A Trace Making Perspective*, edited by Christian Mosbæk and Theo van Leeuwen. New York: Routledge.
Kvåle, Gunhild, and Søren V. Poulsen. Forthcoming. "The Templatized Aesthetics of Wix: A Social Semiotic Technology Approach to Web Design." In *Multimodality and Aesthetics*, edited by Frida Forsgren and Elise Seip Tønnessen, 202–217. London: Routledge.
Langlois, Ganaele. 2014. *Meaning in the Age of Social Media*. New York: Palgrave Macmillan.
Manovich, Lev. 2013. *Software Takes Command*. New York: Bloomsbury.
Martin, James R. 1992. *English Text: System and Structure*. Philadelphia: John Benjamins.
Marino, Mark C. 2006. "Critical Code Studies." *Electronic Book Review*. Accessed 1 June 2018, http://www.electronicbookreview.com/thread/electropoetics/codology.
Leonardi, Paul M. 2010. "Digital Materiality? How Artifacts Without Matter, Matter." *First Monday*. Accessed 31 July 2018, http://firstmonday.org/ojs/index.php/fm/article/view/3036/2567
O'Halloran, Kay L. 2009. "Historical Changes in the Semiotic Landscape: From Calculation to Computation." In *The Routledge Handbook of Multimodal Analysis*, edited by Carey Jewitt, 98–113. London: Routledge.
Page, Ruth, E. David Barton, Johann W. Unger, and Michele Zappavigna. 2014. *Researching Language and Social Media: A Student Guide*. Oxfordshire: Routledge.
Parikka, Jussi. 2012. "New Materialism as Media Theory: Medianatures and Dirty Matter." *Communication and Critical/Cultural Studies* 9 (1): 95–100.
Poulsen, Søren V. 2018a. "Constructing the Corporate Instagram Discourse – A Critical Visual Discourse Approach." *Digital Age in Semiotics and Communication* 1 (1): 95–112.
Poulsen, Søren V. 2018b. "Becoming a Semiotic Technology – A Historical Study of Instagram's Tools for Making and Sharing Photos and Videos." *Internet Histories* 2 (1-2): 121–139.
Poulsen, Søren V. Forthcoming. "Filtered Aesthetics – A Study of Instagram's Photo Filters From a Semiotic Technology Perspective." In *Multimodality and Aesthetics*, edited by Frida Forsgren and Elise Seip Tønnessen, 250–266. London: Routledge.
Roderick, Ian. 2016. *Critical Discourse Studies and Technology: A Multimodal Approach to Analyzing Technoculture*. London: Bloomsbury.
Serazio, Michael, and Brooke E. Duffy. 2018. "Social Media Marketing." In *The SAGE Handbook of Social Media*, edited by Jean Burgess, Alice Marwick, and Thomas Poell, 481–496. London: SAGE.
Van Dijk, Teun A. 2001. "Critical Discourse Analysis." In *The Handbook of Discourse Analysis*, edited by Schiffrin Deborah, Tannen Deborah, and Heidi E. Hamilton, 352–371. London: Blackwell Publishing Ltd.
Van Leeuwen, Theo. 2005. *Introducing Social Semiotics*. London: Routledge.
Van Leeuwen, Theo. 2008a. "New Forms of Writing, New Visual Competencies." *Visual Studies* 23 (2): 130–135.
Van Leeuwen, Theo. 2008b. *Discourse and Practice: New Tools for Critical Discourse Analysis*. Oxford: Oxford University Press.
Van Leeuwen, Theo, Emilia Djonov, and Kay L. O'Halloran. 2013. "'David Byrne Really Does Love PowerPoint': Art as Research on Semiotics and Semiotic Technology." *Social Semiotics* 23 (3): 409–423.
Walton, Marion, Silke Hassreiter, Gary Marsden, and Sena Allen. 2012. "Degrees of Sharing: Proximate Media Sharing and Messaging by Young People in Khayelitsha." Paper presented at ACM SIGCHI's International Conference on Human-Computer Interaction with Mobile Devices and Services (Mobile HCI), San Francisco, September 21–24.
Wodak, Ruth E. 2015. "The Discourse-Historical Approach (DHA)." In *Methods of Critical Discourse Studies*, 3rd ed., edited by Ruth E. Wodak and Micheal Meyer, 23–61. London: Sage.

Zappavigna, Michele. 2012. *Discourse of Twitter and Social Media: How We Use Language to Create Affiliation on the Web*. London: Continuum (now Bloomsbury).
Zappavigna, Michele. 2016. "Social Media Photography: Construing Subjectivity in Instagram Images." *Visual Communication* 15 (3): 271–292.
Zhao, Sumin, and Zappavigna Michele. 2018. "Beyond the Self: Intersubjectivity and the Social Semiotic Interpretation of the Selfie." *New Media & Society* 20 (5): 1735–1754.
Zhao, Sumin, Emilia Djonov, and Theo van Leeuwen. 2014. "Semiotic Technology and Practice: A Multimodal Social Semiotic Approach to PowerPoint." *Text and Talk* 34 (3): 349–375.